

Python Coding Standards: Variables and Functions

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Variable Naming Standards

For Python programs, I will use lowercase names with words separated by underscores, also known as `snake_case`. Variable names should be descriptive so that the purpose of the variable is easy to understand. Examples include `student_name`, `test_score`, and `total_cost`.

I selected this standard because descriptive and consistent variable names make code easier to read, understand, and maintain.

Function Naming and Documentation Standards

For functions, I will use descriptive lowercase names with words separated by underscores. Function names should describe the action performed by the function. For example, `calculate_bmi()` clearly describes what the function does.

I will document functions using Python docstrings. A docstring is a description inside a function that explains what the function does and, when appropriate, its parameters and return value.

```
def calculate_bmi(weight, height):  
    """Calculate and return BMI using weight and height."""
```

I selected these standards because clear function names and documentation help programmers understand and maintain code more easily.

Rationale

I selected these standards because they provide consistency and improve readability. Following commonly accepted Python practices makes programs easier to understand, modify, and maintain. These standards will also help me write clearer programs as I continue learning Python.

Sources

Python Software Foundation. **PEP 8 – Style Guide for Python Code.**

<https://peps.python.org/pep-0008/>

Python Software Foundation. **PEP 257 – Docstring Conventions.**

<https://peps.python.org/pep-0257/>